CIE AS PHYSICS (9702) • MARK SCHEME & MODEL ANSWERS

4.3 Density & Pressure

Syllabus Section: 4. Forces, Density & Pressure **Paper Type:** Multiple Choice (Paper 1) **Total:** 32 Questions (32 Marks)

Question 1 Easy [1 mark]

The formula for hydrostatic pressure is $P = pgh$

Which equation, or principle of physics, is used in the derivation of this formula?

A density = mass ÷ volume

B potential energy = *mgh*

C atmospheric pressure decreases with height

D density increases with depth

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- If you run through the beginning steps of the derivation, you will remember the two equations you start with are
  - $W = mg$
  - $\rho = \frac{m}{V}$
- In an MCQ you would not be required to carry out a derivation, but you will need to know it for the written paper. The full derivation is as follows:

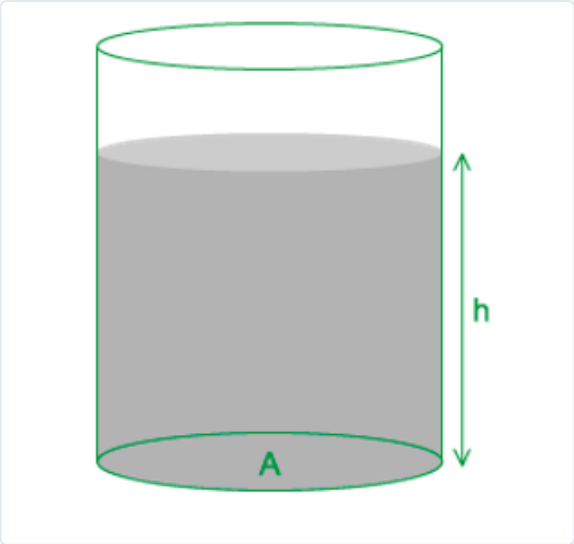
Consider a column of water which occupies a total volume *V*, has a height *h*, and a base surface area of *A*. The weight of all the water is :

$$W = mg$$

The water has a density which is equal to $\rho = \frac{m}{V}$ and therefore $m = \rho V$, hence combining these two equations gives:

$$W = \rho V g$$

For a cylinder, the volume is a product of its base surface area and height:



$$W = \rho Ahg$$

The water has a pressure which is equal to $P = \frac{F}{A}$ and therefore $F = W = PA$, hence combining these t

The water has a pressure which is equal to $P = \frac{F}{A}$ and therefore $F = W = PA$, hence combining these two equations gives:

$$P = \frac{\rho Ahg}{A} = pgh$$

Question 2 Easy [1 mark]

The density of mercury is 13.6×10^3 kg m^{-3}

The pressure difference between the bottom and the top of a column of mercury is 100 kPa.

What is the height of the column?

A 0.75 m

B 1.3 m

C 7.4 m

D 72 m

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- The pressure difference can be established from the equation of hydrostatic pressure
  - $\Delta P = pg\Delta h$
- Density of mercury, ρ = 13 600 kg m^{-3}
- Acceleration due to gravity, g = 9.81 m s^{-2}
- Change in pressure, ΔP = 100 000 Pa
  - $\Delta h = \frac{\Delta P}{pg} = \frac{100\ 000}{13\ 600 \times 9.81} = 0.75$ m
- The height of the column is 0.75 m

Question 3 Easy [1 mark]

An object, immersed in a liquid in a tank, experiences an upthrust.

What is the physical reason for this upthrust?

A the density of the body differs from that of the liquid

B the density of the liquid increases with depth

C the pressure in the liquid increases with depth

D the value of *g* in the liquid increases with depth

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- The physical reason for this upthrust is due to the difference in pressures at the top and the bottom of the object
- If we look at the equation
  - $P = \rho g h$
- We can deduce if the depth *h* increases, the pressure *P* increases too
- Therefore, the pressure at the bottom of the object is greater than the pressure at the top of the object
- The pressure acts on the surface area on the bottom of the object is the upthrust
- Upthrust is a force and this is directly related to pressure by the equation
  - $P = \frac{F}{A}$
- Therefore, the larger the pressure, the larger the force (upthrust) that is exerted on the object

A is incorrect because while this could be true, this is not the physical reason for the upthrust

B & D are incorrect because the density ρ of a liquid remains constant, as does the value of *g* on the Earth's surface

Question 4 Easy [1 mark]

The Mariana Trench in the Pacific Ocean has a depth of about 10 km.

Assuming that sea water is incompressible and has a density of about 1020 kg m^{-3}, what would be the approximate pressure at that depth ?

A 10^5 Pa

B 10^6 Pa

C 10^7 Pa

D 10^8 Pa

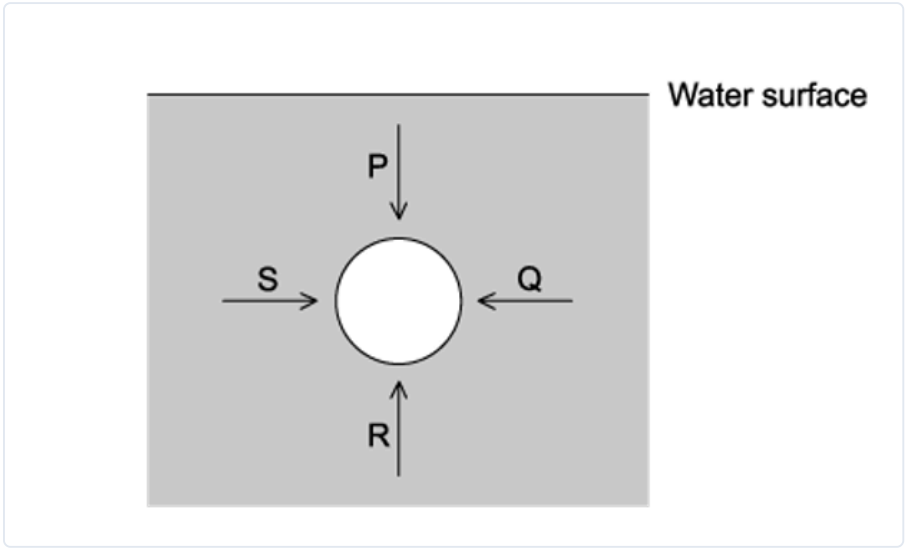
Mark Scheme & Explanation:

*The correct answer is **D** because:*

- The pressure difference can be established from the equation of hydrostatic pressure
  - $\Delta P = pg\Delta h$
- Density of sea water, ρ = 1020 kg m^{-3}
- Acceleration due to gravity, g = 9.81 m s^{-2}
- Depth, Δh = 10 000 m
  - $\Delta P = pg\Delta h = 1020 \times 9.81 \times 10\ \ 000 = 1.0 \times 10^8$ Pa
- The approximate pressure at a depth of 10 km is ~10^8 Pa

Question 5 Easy [1 mark]

The diagram represents a sphere under water. *P*, *Q*, *R* and *S* are forces acting on the sphere, due to the pressure of the water.



Each force acts perpendicularly to the sphere's surface. *P* and *R* act in opposite directions vertically. *Q* and *S* act in opposite directions horizontally.

Which information about the magnitudes of the forces is correct ?

A $P < R$ and $S = Q$

B $P > R$ and $S = Q$

C $P = R$ and $S = Q$ and $P \neq S$

D $P = R$ and $S = Q$ and $P = S$

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- Let's consider the horizontal forces *Q* and *S* initially
- These forces act at the same depth
- Pressure stays the same at the same depth so the magnitude of *Q* and *S* will be the same i.e.
    - $S = Q$
- To determine the relationship between forces *P* and *R* we must look at the equation

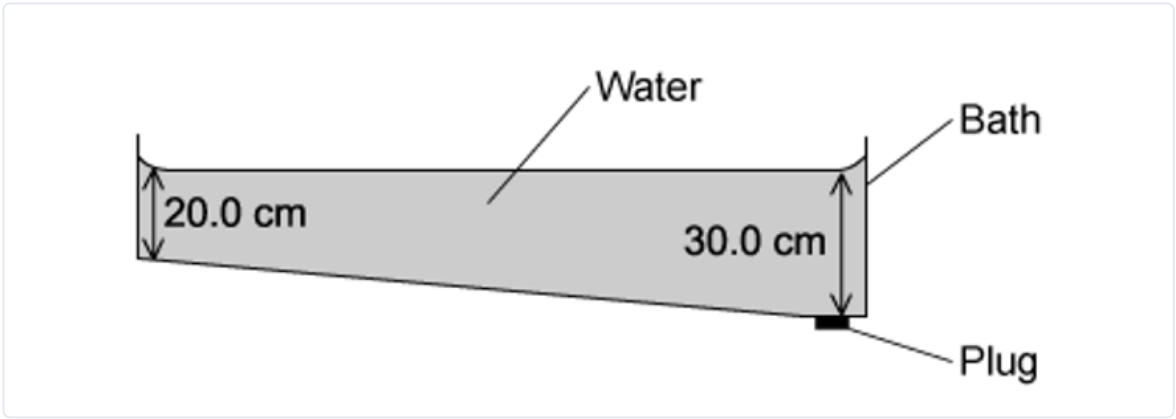
    - $P = pgh$
- We can deduce as the depth *h* increases, the pressure *P* increases too
- Therefore, the pressure at the bottom of the sphere is greater than the pressure at the top of the sphere
- Pressure is directly proportional to force so the force on the bottom of the sphere (*R*) will be greater than the force at the top of the sphere (*P*) i.e.
    - $P < R$

B is incorrect because this implies the pressure is greater nearer the surface of the water which is not correct

C & D are incorrect because these imply there is no difference in pressure at different depth in the water which is also not correct

Question 6 Easy [1 mark]

Water in a bath varies in depth from 20.0 cm at the shallow end to 30.0 cm at the end with the plug.



The density of the water is 1000 kg m^{-3}

What is the pressure of the water acting on the plug ?

A 0.98 kPa

B 2.0 kPa

C 2.9 kPa

D 290 kPa

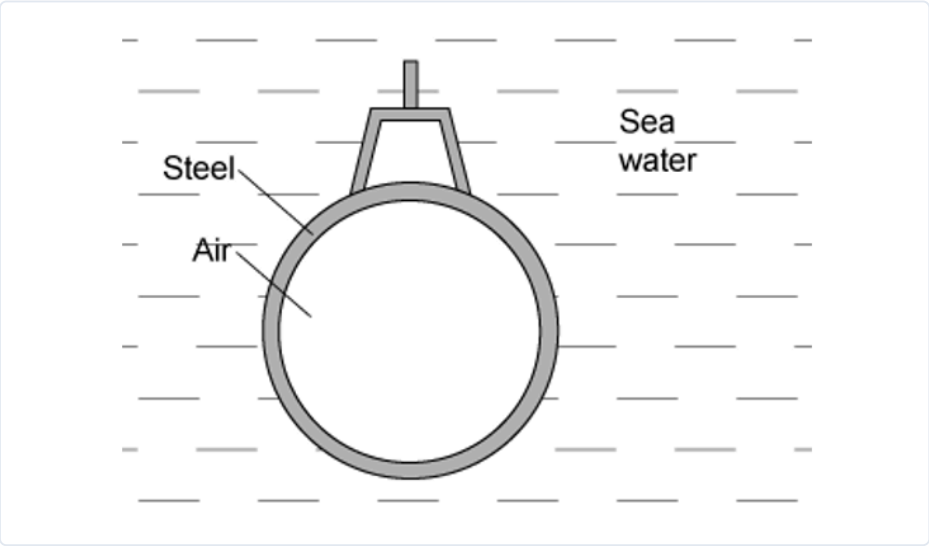
Mark Scheme & Explanation:

*The correct answer is **C** because:*

- The pressure on the plug can be established from the equation of hydrostatic pressure
  - $\Delta P = pg\Delta h$
- Density of water, ρ = 1000 kg m^{-3}
- Acceleration due to gravity, g = 9.81 m s^{-2}
- Depth of plug, Δh = 30 cm = 0.3 m
  - $\Delta P = pg\Delta h = 1000 \times 9.81 \times 0.3 = 2943 \quad \text{Pa}$
- The pressure acting on the plug = 2.9 kPa (2 s. f.)

Question 7 Easy [1 mark]

A submarine is in equilibrium in a fully submerged position.



What causes the upthrust on the submarine?

A the air in the submarine is less dense than sea water

B the sea water exerts a greater upward force on the submarine than the weight of the steel

C the submarine displaces its own volume of sea water

D there is a difference in water pressure acting on the top and on the bottom of the submarine

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- Upthrust occurs because the pressure in a fluid increases with increasing depth
- The pressure acting at the bottom of the submarine is greater than the pressure acting at the top
- Pressure is directly proportional to force, so the force on the bottom of the submarine will be greater than the force at the top of the submarine
- The resultant pressure causes a resultant upward force on the submarine; this is the upthrust

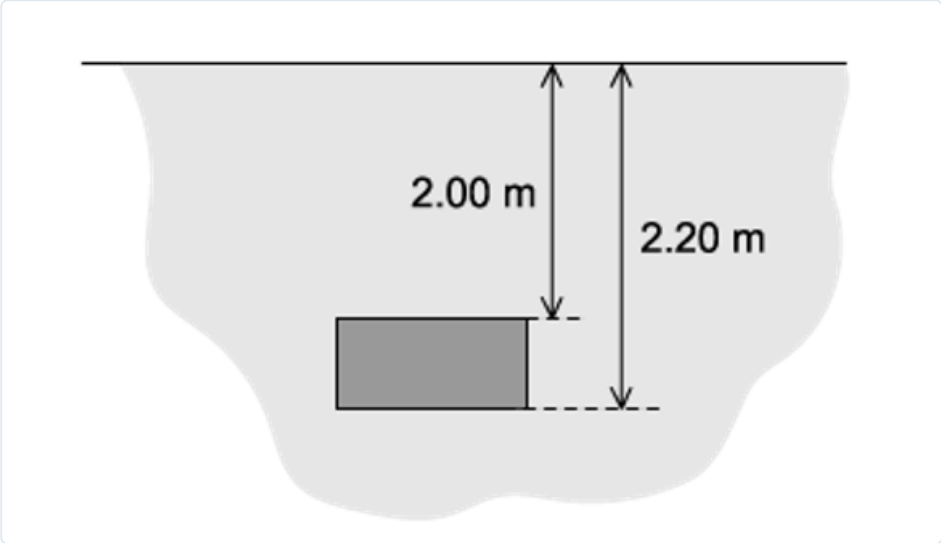
A is incorrect because, although the air inside the submarine is less dense than seawater, the upthrust is produced by the surrounding water, not by the air.

B is incorrect because the submarine is in equilibrium, so the upthrust should be equal to the weight.

C is incorrect because according to Archimedes' principle, the upthrust equals the weight of water displaced and therefore depends on the volume displaced. However, this is a consequence of the pressure difference between top and bottom, not the underlying cause.

Question 8 Easy [1 mark]

The diagram shows a rectangular block of mass 8.2 kg immersed in sea water of density 1.10×10^3 kg m^{-3}



What is the difference in pressure between the top and bottom surfaces of the block?

A 2.2×10^2 Pa

B 2.2×10^3 Pa

C 1.8×10^4 Pa

D 2.3×10^4 Pa

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- The pressure difference can be established from the equation of hydrostatic pressure
  - $\Delta P = \rho g \Delta h$
- Density of sea water, ρ = 1100 kg m^{-3}
- Acceleration due to gravity, g = 9.81 m s^{-2}
- Difference in depth, Δh = 2.2 – 2.0 = 0.2 m
  - $\Delta P = \rho g \Delta h = 1100 \times 9.81 \times 0.2 = 2158 \quad \text{Pa}$

- The pressure difference between the top and bottom = 2.2 × 10^3 Pa (2 s. f.)

Question 9 Easy [1 mark]

A horizontal plate of area 0.036 m^2 is beneath the surface of a liquid of density

930 kg m^{-3}. The force on the plate due to the pressure of the liquid is 290 N.

What is the depth of the plate beneath the surface of the liquid?

A 0.88 m

B 1.13 m

C 8.7 m

D 9.1 m

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- The depth of an object in a liquid can be established from the equation of hydrostatic pressure
  - $P = pgh$
- Pressure is related to force by the equation $P = \frac{F}{A}$
  - $P = pgh = \frac{F}{A}$
  - $h = \frac{F}{pgA}$
- Density of liquid, ρ = 930 kg m^{-3}
- Acceleration due to gravity, g = 9.81 m s^{-2}
- Force on plate, F = 290 N
- Area of plate, A = 0.036 m^2
  - $h = \frac{F}{pgA} = \frac{290}{930 \times 9.81 \times 0.036} = 0.88 \text{ m}$
- The depth of the plate beneath the surface of the liquid is 0.88 m

Question 10 Easy [1 mark]

A rectangular metal bar exerts a pressure of 15 200 Pa on the horizontal surface on which it rests.

If the height of the metal bar is 80 cm, what is the density of the metal ?

A 19 kg m^{-3}

B 190 kg m^{-3}

C 1900 kg m^{-3}

D 19 000 kg m^{-3}

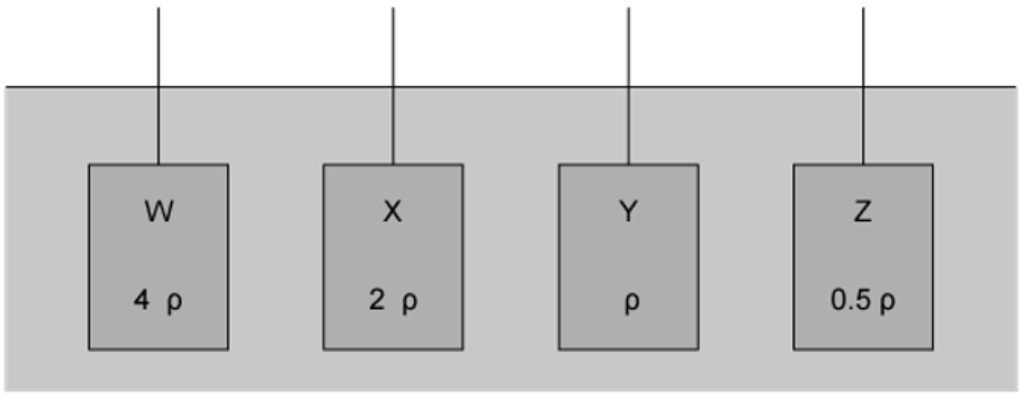
Mark Scheme & Explanation:

*The correct answer is **C** because:*

- The density of an object in a liquid can be determined from the equation of hydrostatic pressure
  - $P = \rho g h$
- Pressure, P = 15 200 Pa
- Acceleration due to gravity, g = 9.81 m s^{-2}
- Height of object, h = 80 cm = 0.8 m
  - $\rho = \frac{P}{gh} = \frac{15\ 200}{9.81 \times 0.8} = 1937 \quad \text{kg} \quad \text{m}^{-3}$
- The density of the metal is 1900 kg m^{-3} (2 s. f.)

Question 11 Medium [1 mark]

Four cuboids with identical length, breadth and height are immersed in water. The cuboids are held at the same depth and in identical orientations by vertical rods, as shown.



Water has density ρ

Cuboid W is made of material of density 4ρ

Cuboid X is made of material of density 2ρ

Cuboid Y is made of material of density ρ

Cuboid Z is made of material of density $\frac{1}{2}\rho$

Which statement is correct?

A the upthrust of the water on each of the cuboids is the same

B the upthrust of the water on W is twice the upthrust of the water on X

C the upthrust of the water on Z is twice the upthrust of the water on Y

D the upthrust of the water on Y is zero

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- Even though the cuboids have different densities, the question states that they are all **held** at the same depth
- Upthrust depends on fluid density and the displaced volume of fluid, not on the object's density

- Since the water density and the cuboid volume are the same for W, X, Y and Z, the upthrust on each is identical, regardless of density

B & **C** are incorrect as upthrust is the same for a given depth and object volume. The difference in density of each cuboid means they have different weights, so if the rods were removed, differences in their motion would be due to differences in weight, not upthrust.

D is incorrect because upthrust will always act on a body submerged in a liquid, this is due to the pressure exerted by the water molecules, so the upthrust on the cuboid cannot be zero while it is submerged.

Question 12 Medium [1 mark]

Atmospheric pressure at sea level has a value of 100 kPa.

The density of sea water is 1020 kg m^{-3}

At which depth in the sea would the total pressure be 150 kPa ?

A 5 m

B 25 m

C 50 m

D 55 m

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- Let the depth in the sea at which pressure is 150 kPa be height *h*
- The pressure at *h* is due to the amount of water above this point and due to the atmospheric pressure at sea level
- The equation of hydrostatic pressure is $P = \rho g h$
- Pressure at depth *h* = pressure due to height *h* of water + atmospheric pressure
  - $150\ \ 000 = \rho g h\ \ +\ \ 100\ \ 000$
  - $\rho g h = 50\ \ 000$
  - $h = \frac{50\ \ 000}{\rho g} = \frac{50\ \ 000}{1020 \times 9.81} = 4.996 = 5.0\ \ \text{m}$
- The pressure is 150 kPa at 5 m below sea level

Question 13 Medium [1 mark]

A beam of α-particles collides with a lead sheet. Each α-particle in the beam has a mass of 6.6×10^{-27} kg and a speed of 1.5×10^{7} m s^{-1}

5.0×10^{4} α-particles per second collide with an area of 1.0 cm^2 of lead. Almost all of the α-particles are absorbed by the lead so that they have zero speed after collision.

What is an estimate of the average pressure exerted on the lead by the α-particles?

A 5.0×10^{-15} Pa

B 5.0×10^{-13} Pa

C 5.0×10^{-11} Pa

D 5.0×10^{-9} Pa

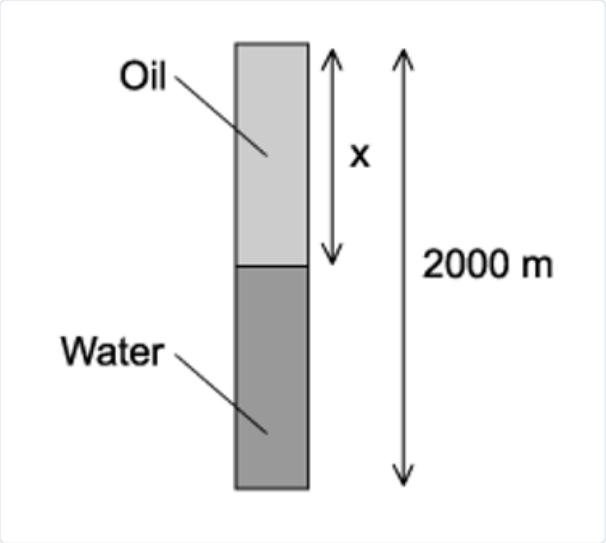
Mark Scheme & Explanation:

*The correct answer is **C** because:*

- Pressure is equal to $P = \frac{F}{A}$
  - Number of particles per second $= \frac{N}{t} = 5.0 \times 10^{4}$
  - Area $= 1.0 \text{ cm}^2 = 1 \times 10^{-4} \text{ m}^2$
  - Mass of one particle $= 6.6 \times 10^{-27} \text{ kg}$
  - Velocity $= 1.5 \times 10^{7} \text{ m s}^{-1}$
- Force is defined as the rate of change of momentum $F = \frac{\Delta p}{\Delta t}$
- Force exerted due to one particle $= \frac{mv}{t}$
- Total force due to N particles $= N\frac{mv}{t} = \frac{N}{t} \times mv$
- Therefore, the average pressure exerted on the lead by the α-particles is:
  - $P = \frac{F}{A} = \frac{N}{t} \times \frac{mv}{A}$
  - $P = \frac{5.0 \times 10^{4} \times 6.6 \times 10^{-27} \times 1.5 \times 10^{7}}{1 \times 10^{-4}}$
  - $P = 4.95 \times 10^{-11} \approx 5.0 \times 10^{-11} \text{ Pa}$

Question 14 Medium [1 mark]

A bore-hole of depth 2000 m contains both oil and water as shown. The pressure due to the liquids at the bottom of the bore-hole is 17.5 MPa. The density of the oil is 830 kg m^{-3} and the density of the water is 1000 kg m^{-3}



What is the depth *x* of the oil ?

A 907 m

B 1000 m

C 1090 m

D 1270 m

Mark Scheme & Explanation:

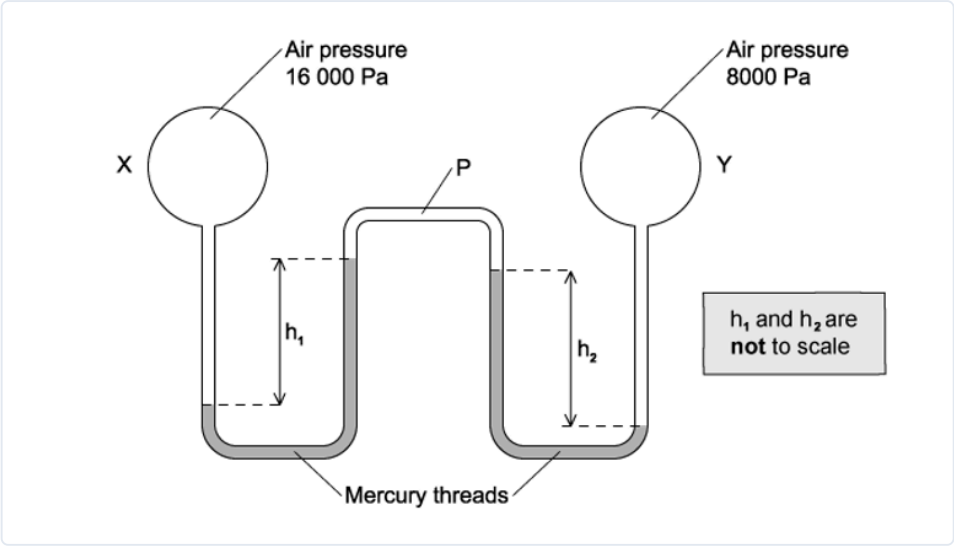
*The correct answer is **D** because:*

- The pressure at the bottom of the bore-hole is the sum of the pressures exerted by water and oil on the bottom which is equal to 17.5 MPa
- The equation of hydrostatic pressure is $P = \rho g h$
- For the column of water:
  - Depth of water, $h = 2000 - x$
  - Density of water, $\rho = 1000 \quad \text{kg} \quad \text{m}^{-3}$
  - Acceleration due to gravity, $g = 9.81 \quad \text{m} \quad \text{s}^{-2}$

- Pressure exerted by water $= 2000 - x 10009.81$
- For the column of oil:
  - Depth of oil, $h = x$
  - Density of oil, $\rho = 830 \quad \text{kg} \quad \text{m}^{-3}$
  - Pressure exerted by oil $= 8309.81x$
- The sum of the pressures is 17.5 MPa
  - $2000 - x 10009.81 + 8309.81x = 17.5 \times 10^{6}$
  - $2000 \times 1000 \times 9.81 - 17.5 \times 10^{6} = 1000 \quad gx - 830 \quad gx$
  - $2.12 \times 10^{6} = 170 \quad gx$
  - $x = \frac{2.12 \times 10^{6}}{170 \times 9.81} = 1271 \quad \text{m}$
- The depth of oil in the bore-hole is about 1270 m

Question 15 Medium [1 mark]

Two bulbs X and Y containing air at different pressures are connected by a tube P which contains two mercury threads.



The density of mercury is 13 600 kg m^{-3}

Which pair of values of h_1 and h_2 is possible?

	h_1 / cm	h_2 / cm
A	4.0	2.0
B	6.0	6.0
C	12.0	18.0
D	18.0	12.0

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- The equation of hydrostatic pressure is $P = \rho g h$
- Let the pressure at P = p

- Let's consider the two sides separately
- On the left section:
  - $16\ \ 000 - p = h_1 \rho g$ **(1)**
- On the right section:
  - $8000 - p = h_2\ \ \rho g$ **(2)**
- If we take equation **(2)** from equation **(1)**
  - $16\ \ 000 - 8000 = h_1 - h_2\ \ \rho g$
  - $h_1 - h_2 = \frac{8000}{\rho g} = \frac{8000}{13\ \ 600 \times 9.81} = 0.06\ \ \text{m}$
- The difference between heights and is 0.06 m or 6 cm. The only set of values which matches this difference is choice D where $h_1 = 18.0\ \ \text{cm}$ and $h_2 = 12.0\ \ \text{cm}$

Question 16 Medium [1 mark]

1.5 m^3 of water is mixed with 0.50 m^3 of alcohol. The density of water is 1000 kg m^{-3} and the density of alcohol is 800 kg m^{-3}

What is the density of the mixture with volume 2.0 m^3 ?

A 850 kg m^{-3}

B 900 kg m^{-3}

C 940 kg m^{-3}

D 950 kg m^{-3}

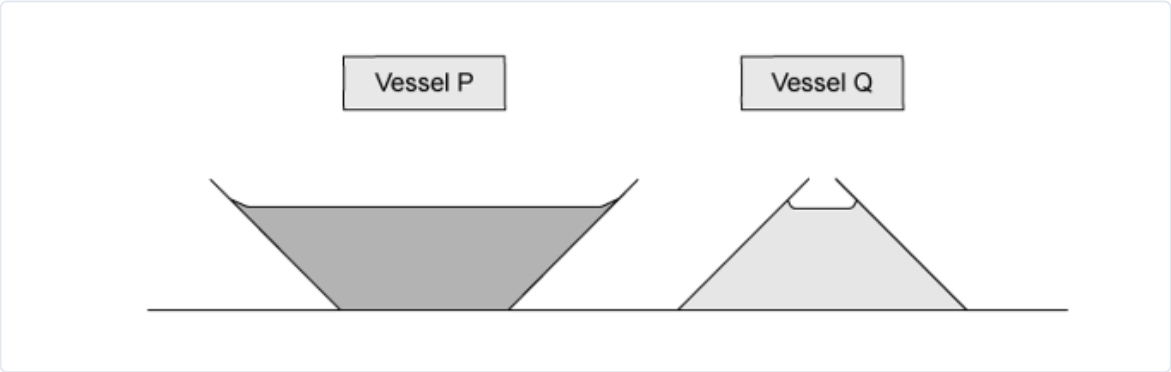
Mark Scheme & Explanation:

*The correct answer is **D** because:*

- Density of mixture $= \frac{total\ \ mass}{total\ \ volume}$
- Total volume = 2.0 m^3
- For water:
  - Density, $\rho = 1000\ \ \text{kg}\ \ \text{m}^{-3}$
  - Volume, $V = 1.5\ \ \text{m}^3$
  - Mass, $m = \rho V = 1000 \times 1.5 = 1500\ \ \text{kg}$
- For alcohol:
  - Density, $\rho = 800\ \ \text{kg}\ \ \text{m}^{-3}$
  - Volume, $V = 0.5\ \ \text{m}^3$
  - Mass, $m = \rho V = 800 \times 0.5 = 400\ \ \text{kg}$
- Total mass $= 1500 + 400 = 1900\ \ \text{kg}$
- Density of mixture $= \frac{total\ \ mass}{total\ \ volume} = \frac{1900}{2} = 950\ \ \text{kg}\ \ \text{m}^{-3}$
- The density of the mixture is 950 kg m^{-3}

Question 17 Medium [1 mark]

The diagram shows two vessels, P and Q, both with sides inclined at 45°



Vessel P tapers outwards and vessel Q tapers inwards, as shown.

Both vessels contain a liquid. The depth of the liquid in the vessels is the same. The liquid in vessel P is twice as dense as the liquid in vessel Q.

What is the ratio $\frac{pressure\ due\ to\ the\ liquid\ on\ the\ base\ of\ P}{pressure\ due\ to\ the\ liquid\ on\ the\ base\ of\ Q}$?

A $\frac{2}{1}$

B $\frac{\sqrt{2}}{1}$

C $\frac{1}{\sqrt{2}}$

D $\frac{1}{2}$

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- The equation of hydrostatic pressure is $P = \rho g h$
- The pressure *P* depends only on the depth h, the density ρ of the liquid and the acceleration of free fall *g*. The volume of the container does not affect the pressure at any point
- Let the density of the liquid in vessel Q = ρ and the density of the liquid in vessel P = 2ρ
- The depth *h* of the liquids is the same
- For vessel P:
  - $P = h2\rho g = 2\rho g h$
- For vessel Q:

- $P = \rho g h$

- Ratio: $\frac{pressure\ \ due\ \ to\ \ the\ \ liquid\ \ on\ \ the\ \ base\ \ of\ \ P}{pressure\ \ due\ \ to\ \ the\ \ liquid\ \ on\ \ the\ \ base\ \ of\ \ Q} = \frac{2\rho g h}{\rho g h} = \frac{2}{1}$

Question 18 Medium [1 mark]

At a depth of 20 cm in a liquid of density 1800 kg m^{-3}, the pressure due to the liquid is *p*.

Another liquid has a density of 1200 kg m^{-3}

What is the pressure due to this liquid at a depth of 60 cm ?

A $\frac{p}{2}$

B $\frac{3p}{2}$

C $2p$

D $3p$

Mark Scheme & Explanation:

The correct answer is ***C*** *because:*

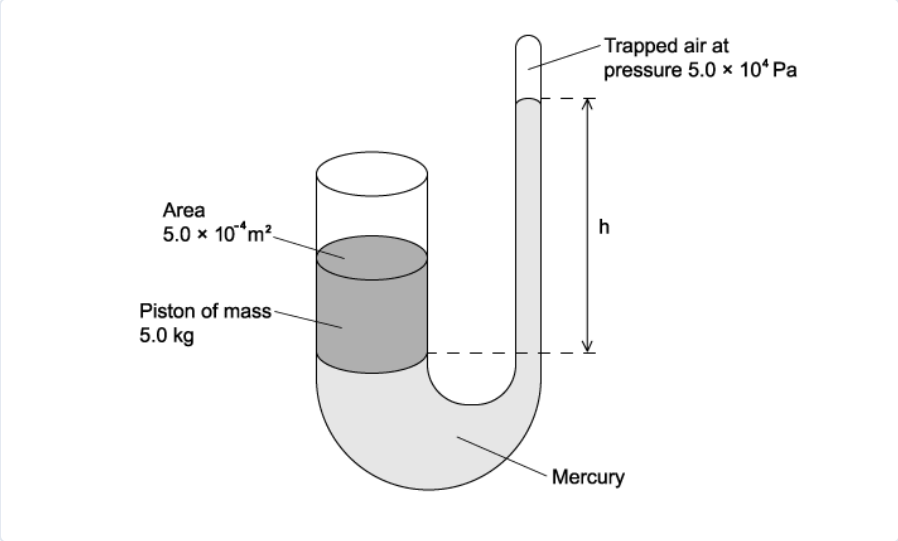
- The equation of hydrostatic pressure is $P = \rho g h$
- Pressure of liquid 1, P_1:
  - Pressure, $P_1 = p$
  - Depth of water, $h = 20 \text{ cm} = 0.2 \text{ m}$
  - Density of water, $\rho = 1800 \text{ kg m}^{-3}$
  - Acceleration due to gravity, $g = 9.81 \text{ m s}^{-2}$
- $P_1 = \rho g h_1 = 1800 \times 9.81 \times 0.2$
- Pressure of liquid 2, P_2:
  - Depth of water, $h = 60 \text{ cm} = 0.6 \text{ m}$
  - Density of water, $\rho = 1200 \text{ kg m}^{-3}$
- $P_2 = \rho_2 g h_2 = 1200 \times 9.81 \times 0.6$
- $\frac{P_1}{P_2} = \frac{\rho_1 g h_1}{\rho_2 g h_2} = \frac{1800 \times 9.81 \times 0.2}{1200 \times 9.81 \times 0.6} = \frac{360}{720} = \frac{1}{2}$
- $\frac{p}{P_2} = \frac{1}{2}$
- $P_2 = 2p$

- The pressure of liquid 2 at 60 cm is two times larger than the pressure of liquid 1 at 20 cm

Question 19 Medium [1 mark]

A U-tube closed at one end contains mercury. Air at a pressure of 5.0×10^4 Pa is trapped at the closed end. The other end is open to the atmosphere and is fitted with a piston of mass 5.0 kg and cross-sectional area 5.0×10^{-4} m^2

The density of mercury is 13 600 kg m^{-3} and atmospheric pressure is 1.01×10^5 Pa



What is the height of the mercury column?

A 37 cm

B 44 cm

C 74 cm

D 110 cm

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- The equation of hydrostatic pressure is $P = \rho g h$
- At the same level in the liquid mercury, the pressure on both sides of the U-tube should be the same
- Consider the pressure at the lower level indicated - where the piston is in contact with the liquid mercury
- Pressure at the left end = Pressure at the right end
- The pressure on the left is due to the weight of the piston and due to atmospheric pressure above the piston

- The pressure on the right is due to the height *h* of the column of mercury and due to the trapped air above the mercury
- The pressure of the trapped air plus the pressure of the mercury column of height *h* must equal total pressure supplied by the piston and the atmosphere
- Pressure due to piston $= \frac{\text{force}}{\text{area}} = \frac{\text{weight}}{\text{area}} = \frac{mg}{A}$
- Pressure at left end = Pressure at right end
- Pressure due to piston and atmosphere = Pressure due to mercury column and trapped air
  - $\frac{mg}{A} + P_{atm} = \rho g h + P_{air}$
  - $\frac{5.0 \times 9.81}{5.0 \times 10^{-4}} + 1.01 \times 10^{5} = h \times 13 \quad 600 \times 9.81 + 5.0 \times 10^{4}$
  - $9.81 \times 10^{4} + 1.01 \times 10^{5} - 5.0 \times 10^{4} = 133 \quad 416 \quad h$
  - $149 \quad 100 = 133 \quad 416 \quad h$
  - $h = \frac{149 \quad 100}{133 \quad 416} = 1.1 \quad \text{m}$
- Therefore, the height of the mercury column is 110 cm

Question 20 Medium [1 mark]

A ball is falling at terminal speed in still air. The forces acting on the ball are upthrust, viscous drag and weight.

What is the order of increasing magnitude of these three forces?

A upthrust → viscous drag → weight

B viscous drag → upthrust → weight

C viscous drag → weight → upthrust

D weight → upthrust → viscous drag

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- At terminal speed, the resultant force on the ball is zero. As weight is the only downward force, it must be the largest of the three
- The upthrust is a result of the force due to the difference in pressure at the top and bottom of the ball
    - Pressure, $P = \rho g h$
    - Force, $F = PA$
- Air is not very dense (low ρ) and the difference in height h from the top of the ball to its bottom is also very small. Thus, the upthrust force is small
- Viscous drag (or air resistance), on the other hand, will almost be as large as the weight when a body is travelling at terminal velocity
- So, from smallest to largest force: upthrust → viscous drag → weight

It is a common misconception that the Archimedean upthrust on a body in air is quite large, however, this is incorrect, and it is the opposite which is true. The upthrust will only be about a thousandth or less than the weight of the solid

Question 21 Medium [1 mark]

A submarine is designed to operate at various depths in the ocean.

The atmospheric pressure at the ocean surface is 101 kPa.

The density of ocean water is 1025 kg m^{-3}.

At what depth would the absolute pressure on the submarine's hull be 202 kPa?

A 10 m

B 20 m

C 30 m

D 40 m

Mark Scheme & Explanation:

The correct answer is A

- The absolute pressure at depth consists of atmospheric pressure plus hydrostatic pressure
- Using the hydrostatic pressure equation:

$$P_{total} = P_{atmospheric} + \rho g h$$

- Substituting known values

$$202\ \ 000 = 101\ \ 000 + (1025)(9.81)h$$

- Rearranging

$$101\ \ 000 = (1025)(9.81)h$$

- Solving for depth

$$h = \frac{101\ \ 000}{1025 \times 9.81} = \frac{101\ \ 000}{10\ \ 055} = 10.0\ \ \text{m}$$

B is incorrect as 20 m would give a total pressure of approximately 303 kPa

C is incorrect as 30 m would give a total pressure of approximately 404 kPa

D is incorrect as 40 m would give a total pressure of approximately 505 kPa

Question 22 Hard [1 mark]

The density of water is 1.0 g cm^{-3} and the density of glycerine is 1.3 g cm^{-3}

Water is added to a measuring cylinder containing 40 cm^3 of glycerine so that the density of the mixture is 1.1 g cm^{-3}. Assume that the mixing process does not change the total volume of the liquid

What is the volume of water added ?

A 40 cm^3

B 44 cm^3

C 52 cm^3

D 80 cm^3

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- When the water and the glycerin are added together, the density of the mixture is obtained using $\frac{\text{total mass}}{\text{total volume}}$
- For water:
  - Density of water, $\rho_w = 1.0 \quad g \quad \text{cm}^{-3}$
  - Mass of water $= m_w$
  - Volume of water $= V_w$
- For glycerine:
  - Density of glycerine, $\rho_g = 1.3 \quad g \quad \text{cm}^{-3}$
  - Mass of glycerine $= m_g$
  - Volume of glycerine, $V_g = 40 \quad \text{cm}^3$
- The mass of the glycerine can be found using
  - Mass, $m_g = \rho_g \times V_g = 1.3 \times 40 = 52 \quad \text{g}$
- Consider the mixture:
  - Density of mixture, $\rho = 1.1 \quad \text{g} \quad \text{cm}^{-3}$
  - Total mass of mixture $= m_w + m_g = m_w + 52$
- The mixing process does not change the total volume of the liquid

- Total volume of mixture $= V_w + 40$

- Density of mixture $= \frac{\text{total mass}}{\text{total volume}}$
  - $1.1 = \frac{m_w + 52}{V_w + 40}$
  - $V_w + 40 = \frac{m_w + 52}{1.1}$ **(1)**

- Consider the density of water
  - $m_w = \rho_w \times V_w = 1.0 \quad V_w = V_w$ **(2)**

- Replace m_w by V_w in equation **(1)**
  - $V_w + 40 = \frac{V_w + 52}{1.1}$
  - $V_w = \frac{V_w}{1.1} + \frac{52}{1.1} - 40$
  - $V_w - \frac{V_w}{1.1} = \frac{52}{1.1} - 40$
  - $V_w 1 - \frac{1}{1.1} = \frac{80}{11}$
  - $V_w = 80 \quad \text{cm}^3$

Question 23 Hard [1 mark]

A cylindrical block of wood has cross-sectional area *A* and weight *W*. It is totally immersed in water with its axis vertical. The block experiences pressures p_t and p_b at its top and bottom surfaces respectively.

Which expression is equal to the upthrust on the block ?

A $p_b - p_t A + W$

B $p_b - p_t$

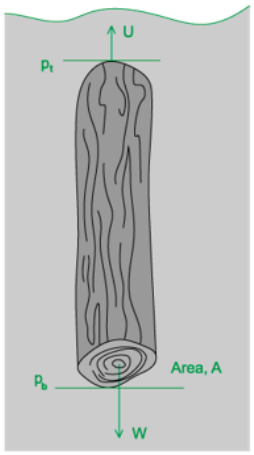
C $p_b - p_t A$

D $p_b - p_t A - W$

Mark Scheme & Explanation:

The correct answer is ***C*** *because:*

- The upthrust of the block is the net vertical force acting on the block
- Pressure is equal to $p = \frac{F}{A}$
- Therefore, force is equal to $F = p \times A$ (where A is the cross-sectional area)
- Now, because the bottom of the block is deeper into the water than the top of the block, $p_b > p_t$
- So, the force due to the pressure at the bottom of the block $= p_b \times A$
- And the force due to the pressure at the top of the block $= p_t \times A$
- Upthrust on block $= p_b A - p_t A = p_b - p_t \quad A$



A & D are incorrect because these expressions correctly (D) and incorrectly (A) describe the resultant force on the wooden cylinder

B is incorrect because this expression describes the difference in pressure between the top and the bottom of the wooden cylinder

Question 24 Hard [1 mark]

Full-fat milk is made up of fat-free milk mixed with fat.

A volume of 1.000×10^{-3} m^3 of full-fat milk has a mass of 1.035 kg. It contains 4.00% fat by volume.

The density of fat-free milk is 1.040×10^3 kg m^{-3}

What is the density of fat ?

A 1.25×10^2 kg m^{-3}

B 9.15×10^2 kg m^{-3}

C 9.28×10^2 kg m^{-3}

D 1.16×10^3 kg m^{-3}

Mark Scheme & Explanation:

*The correct answer is **B** because:*

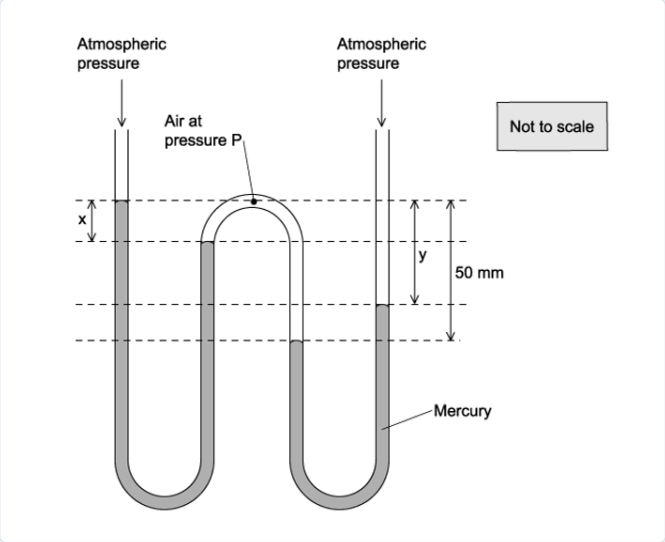
- Density of full-fat milk $= \frac{\text{total mass}}{\text{total volume}}$
- Total mass = mass of fat + mass of fat-free milk
- Total volume = volume of fat + volume of fat-free milk
- Consider the full-fat milk
  - Mass of full-fat milk $= 1.035$ kg
  - Volume of full-fat milk $= 1.000 \times 10^{-3}$ m^3
  - Density of full-fat milk $= \frac{\text{mass}}{\text{volume}} = \frac{1.035}{1.000 \times 10^{-3}} = 1035$ kg m^{-3}
- The full-fat milk contains 4.00% fat by volume
  - Volume of fat $= 0.04 \times 1.000 \times 10^{-3} = 4.0 \times 10^{-5}$ m^3
- Therefore, the fat-free milk makes up the remaining 96% of the volume
  - Volume of fat-free milk $= 0.96 \times 1.000 \times 10^{-3} = 9.6 \times 10^{-4}$ m^3
- Given that the density of fat-free milk $= 1.040 \times 10^3$ kg m^{-3}
  - Mass of fat-free milk $=$ density $\times$ volume
  - Mass of fat-free milk $= 1.040 \times 10^3 \times 9.6 \times 10^{-4} = 0.9984$ kg
- Total mass = mass of fat + mass of fat-free milk

- Mass of fat $= 1.035 - 0.9984 = 0.0366 \quad \text{kg}$
    - Density of fat $= \frac{\text{mass}}{\text{volume}} = \frac{0.0366}{4.0 \times 10^{-5}} = 915 \quad \text{kg} \quad \text{m}^{-3}$
- The density of fat is therefore equal to 9.15×10^2 kg m^{-3}

The question indicates that up to four significant figures should be considered as quantities such as volume are given as 1.000×10^{-3} rather than 1.0×10^{-3}, not doing so could affect the final answer you obtain.

Question 25 Hard [1 mark]

A W-shaped tube contains two amounts of mercury, each open to the atmosphere. Air at pressure *P* is trapped in between them. The diagram shows two vertical distances *x* and *y*.



Atmospheric pressure is equal to the pressure that would be exerted by a column of mercury of height 760 mm. The pressure *P* is expressed in this way.

Which values of *x*, *y* and *P* are possible ?

	***x* / mm**	***y* / mm**	***P* / mm of Mercury**
A	20	20	780
B	20	30	780
C	30	20	810
D	30	30	790

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **B** because:*

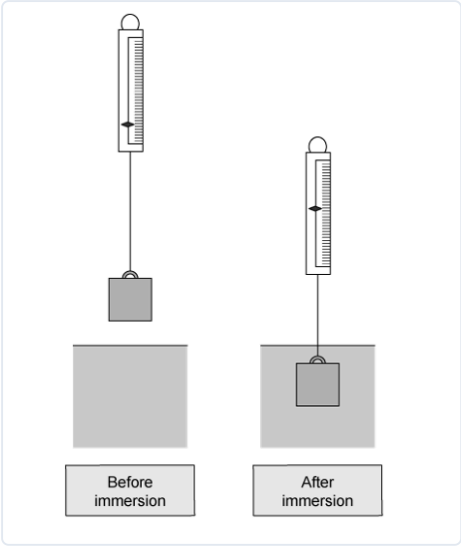
- Atmospheric pressure P_A = column of mercury of height 760mm

- Compare the left part with the middle (left) part
- Since the position of the mercury column is lower in the middle, the pressure P is greater than the atmospheric pressure acting at the left 'open' tube
  - $P - P_A = x$
  - $P - 760 = x$ **(1)**
- Now, compare the right part of the diagram with the middle (right) part
  - $P - P_A = 50 - y$
  - $P - 760 = 50 - y$
  - $P + y = 810$ **(2)**
- The values from the table should satisfy both equation **(1)** and **(2)**
- Consider option B:
  - x = 20, P = 780
- Equation **(1)**: $780 - 760 = 20$
  - y = 30, P = 780
- Equation **(2)**: $780 + 30 = 810$
- These values satisfy equations **(1)** and **(2)** so option B is correct

A is incorrect because	x = 20, P = 780 Equation **(1)**: $780 - 760 = 20$ ✓ y = 20, P = 780 Equation **(2)**: $780 + 20 = 810$ ×
C is incorrect because	x = 30, P = 810 Equation **(1)**: $810 - 760 = 30$ × y = 20, P = 810 Equation **(2)**: $810 + 20 = 810$ ×
D is incorrect because	x = 30, P = 790 Equation **(1)**: $790 - 760 = 30$ ✓ y = 30, P = 790 Equation **(2)**: $790 + 30 = 810$ ×

Question 26 Hard [1 mark]

The diagram shows a metal cube of density ρ suspended from a spring balance before and during immersion in water. Each face of the cube has a surface area A and each edge has length h.



A reduction in the balance reading occurs as a consequence of the immersion. The cube experiences pressures p_1 and p_2 at its top and bottom surfaces respectively.

What is the value of the upthrust on the metal cube during immersion ?

A $Ah\rho g$

B $Ah\rho g + p_1 A$

C $p_2 A$

D $p_2 A - p_1 A$

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- The upthrust of the block is the net vertical force acting on the block
- Pressure is equal to $p = \frac{F}{A}$
- Therefore, force is equal to $F = p \times A$ (where A is the cross-sectional area)
- Now, because the bottom of the block is deeper into the water than the top of the block, $p_2 > p_1$
- So, the force due to the pressure at the bottom of the block $= p_2 \times A$

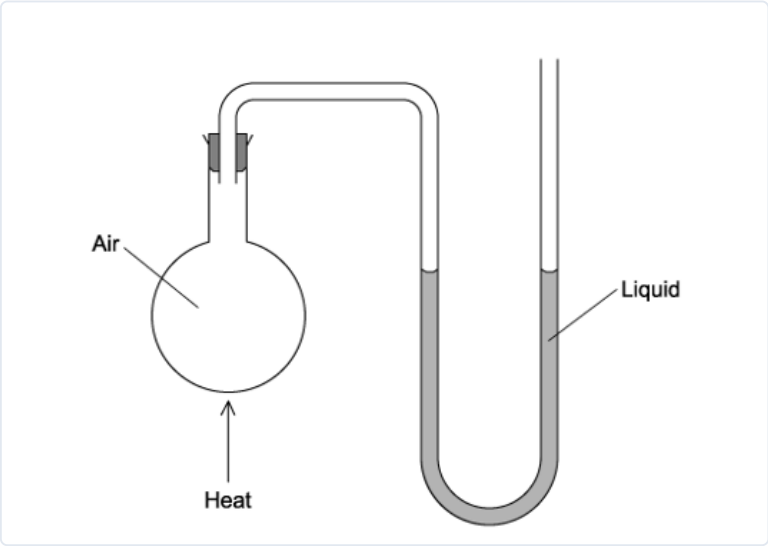
- And the force due to the pressure at the top of the block $= p_1 \times A$
- Upthrust on block $= p_2 A$ - $p_1 A$

A & B are incorrect because the term $Ah\rho g$ describes the force on the cube at a certain depth under the water, the value of this changes depending on the depth of the cube but upthrust stays the same, no matter the depth

C is incorrect because the term $p_2 A$ describes the force on the bottom surface of the cube, not the upthrust as, again, the value of this changes depending on the depth due to the pressure change

Question 27 Hard [1 mark]

The diagram shows a flask connected to a U-tube containing liquid. The flask contains air at atmospheric pressure.



The flask is now gently heated and the liquid level in the right-hand side of the U-tube rises through a distance *h*. The density of the liquid is *ρ*.

What is the increase in pressure of the heated air in the flask ?

A $h\rho$

B $\frac{1}{2}h\rho g$

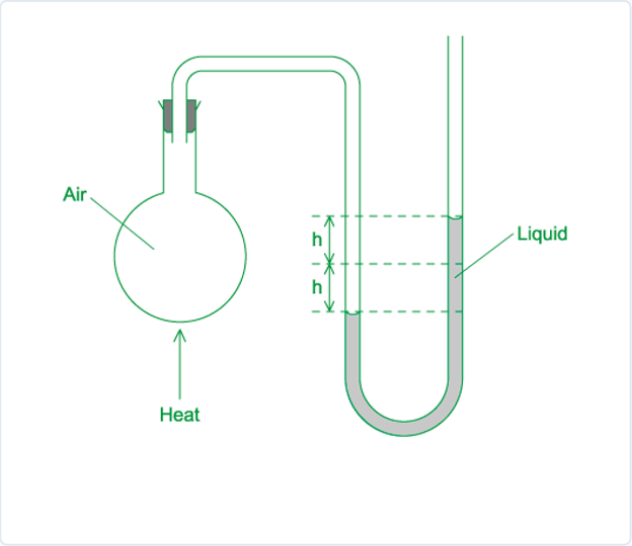
C $h\rho g$

D $2h\rho g$

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- The equation of hydrostatic pressure is $\Delta P = \Delta h\rho g$
- The heating causes the liquid in the right-hand side to rise a distance *h* from its original position
- But this also causes the liquid in the left-hand side to sink by a distance *h* from its original position
- So, the difference in height of the liquid is $\Delta h = h + h = 2h$
- Therefore, the increase in pressure is
    - $\Delta P = \Delta h\rho g = 2h\rho g$


Air
Heat
h
h
Liquid

Question 28 Hard [1 mark]

Two solid substances P and Q have atoms of mass M_P and M_Q respectively. They have n_P and n_Q atoms per unit volume.

The density of P is greater than the density of Q.

Which expression **must** be correct ?

A $M_P > M_Q$

B $n_P > n_Q$

C $M_{:P}\, n_P > M_Q\, n_Q$

D $\frac{M_P}{n_P} > \frac{M_Q}{n_Q}$

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- The mass of one atom of P and Q are M_P and M_Q respectively
- The number of atoms per unit volume of P and Q are n_P and n_Q respectively
- Total mass per unit volume $= Mn$
- Density $= \frac{total\ \ mass}{volume}$
  - Density of P $= M_P n_P$
  - Density of Q $= M_Q n_Q$
- The density of P is greater than the density of Q so
  - $M_P n_P > M_Q n_Q$

A & B are incorrect because it cannot be said conclusively whether or not $M_P > M_Q$ or $n_P > n_Q$ is true because all we know is that the density of P is greater than the density of Q so we cannot conclude the individual values of *M* and *n* from this information alone

D is incorrect because n represents the number of atoms per unit volume so $\frac{M}{n}$ is equivalent to $mass \times volume$, which is an incorrect expression for density

Question 29 Hard [1 mark]

Liquid Q has twice the density of liquid R.

At depth *x* in liquid R, the pressure due to the liquid is 4 kPa.

At what depth in liquid Q is the pressure due to the liquid 7 kPa ?

A $\frac{2x}{7}$

B $\frac{7x}{8}$

C $\frac{8x}{7}$

D $\frac{7x}{2}$

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- The equation of hydrostatic pressure is $P = h\rho g$
- For liquid R:
  - Pressure, $P_R = h\rho_R g = x\rho_R g = 4$
  - Density, $\rho_R = \frac{4}{xg}$ **(1)**
- For liquid Q:
  - Density, $\rho_Q = 2\rho_R$
  - Pressure, $P_Q = h\rho_Q g = h2\rho_R g = 7$
- So, depth in liquid Q:
  - $h = \frac{7}{2\rho_R g}$ **(2)**
- Substituting equation **(1)** into **(2)**:
  - $h = \frac{7}{2g\frac{4}{xg}} = \frac{7}{2}\frac{4}{x}^{-1} = \frac{7x}{8}$

Exam Tip: in questions such as these, identify the link between the two quantities in order to equate them. For example, in this question, ρ is the quantity that links R and Q, so identifying ρ in terms of the other known variables is a crucial step

Question 30 Hard [1 mark]

A child drinks a liquid of density ρ through a vertical straw.

Atmospheric pressure is p_0 and the child is capable of lowering the pressure at the top of the straw by 10%. The acceleration of free fall is g.

What is the maximum length of straw that would enable the child to drink the liquid?

A $\frac{p_0}{10\rho g}$

B $\frac{9p_0}{10\rho g}$

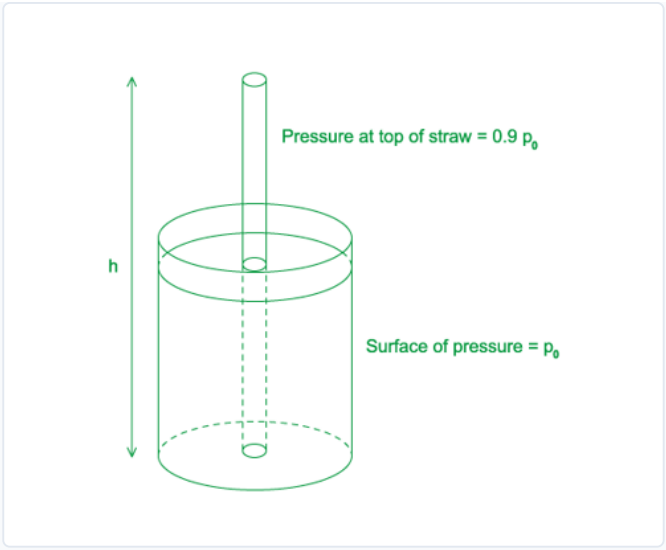
C $\frac{p_0}{\rho g}$

D $\frac{10p_0}{\rho g}$

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- The equation of hydrostatic pressure is $P = h\rho g$
- Imagine a beaker containing the liquid and a vertical straw (not inclined)
- Atmospheric pressure p_0 acts on the surface of the liquid
- The child is capable of lowering the pressure at the top of the straw by 10%
- So, the child is drinking from the top of the straw with a pressure of $1 - 0.1p_0 = 0.9p_0$
- As the child drinks, the liquid rises up the straw of length h
- The difference in pressure at the surface if the liquid in the beaker p_0 and the pressure at the top of the straw $0.9p_0$ is equal to the pressure due to the column of liquid in the straw (from the equation of hydrostatic pressure)
  - $p_0 - 0.9p_0 = h\rho g$
  - $h\rho g = 0.1p_0 = \frac{p_0}{10}$
  - $h = \frac{p_0}{10\rho g}$
- The length of the straw is equal to $h = \frac{p_0}{10\ \rho g}$


Pressure at top of straw = 0.9 p_0
h
Surface of pressure = p_0

Question 31 Hard [1 mark]

Icebergs typically float with a large volume of ice beneath the water. Ice has a density of 917 kg m^{-3} and a volume

The density of seawater is 1020 kg m^{-3}

What fraction of the ice is submerged underwater ?

A $0.05 \quad V_i$

B $0.10 \quad V_i$

C $0.90 \quad V_i$

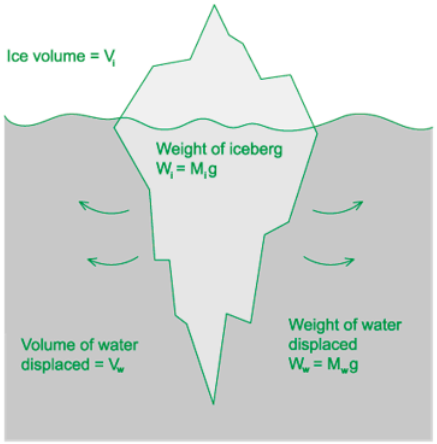
D $0.95 \quad V_i$

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- According to Archimedes' principle, the buoyant force on an object equals the weight of the fluid it displaces
- The iceberg has weight $W_i = M_i g$ and the buoyant force is equal to the weight of the displaced water, $W_w = M_w g$
- Furthermore, since the iceberg is floating, its weight exactly balances the buoyant force:
  - $W_w = W_i$
  - $M_w \quad g = M_i \quad g$
- Density is equal to $\frac{mass}{volume}$ so mass = density × volume
  - $V_w \quad \rho_w \quad g = V_i \quad \rho_i \quad g$
  - $V_w = \frac{\rho_i}{\rho_w} V_i$
- So, the fraction of ice underwater $\frac{V_w}{V_i}$ is given by the ratio of densities
  - $\frac{V_w}{V_i} = \frac{\rho_i}{\rho_w} = \frac{917}{1020} = 0.899$
  - $V_w = 0.90 \quad V_i$

Therefore, 90% of an iceberg's volume (and mass) is submerged underwater


Ice volume = V_i
Weight of iceberg
$W_i = M_i g$
Volume of water
displaced = V_w
Weight of water
displaced
$W_w = M_w g$

Question 32 Hard [1 mark]

A wooden block is floating in a tank of oil. The density of the wood is $650\ \text{kg m}^{-3}$ and the density of the oil is $900\ \text{kg m}^{-3}$.

What volume of the wooden block is submerged in the oil?

A $0.28V_{\text{block}}$

B $0.50V_{\text{block}}$

C $0.72V_{\text{block}}$

D $0.90V_{\text{block}}$

Mark Scheme & Explanation:

The correct answer is C

- According to Archimedes' principle, the buoyant force (upthrust) on an object equals the weight of the fluid it displaces

$$F_{upthrust} = W_{oil}$$

- The block floats, so the buoyant force equals the weight of the block

$$F_{upthrust} = W_{block}$$

- Therefore

$$W_{oil} = W_{block}$$

$$m_{oil}\,g = m_{block}\,g$$

- Density is equal to $\frac{\text{mass}}{\text{volume}}$ so mass = density × volume

$$V_{oil}\rho_{oil}\,g = V_{block}\rho_{block}\,g$$

$$V_{oil}\rho_{oil} = V_{block}\rho_{block}$$

$$V_{oil} = \frac{\rho_{block}}{\rho_{oil}}V_{block}$$

- Therefore, the fraction of the block submerged $\frac{V_{oil}}{V_{block}}$ is given by the ratio of densities

$$\frac{V_{oil}}{V_{block}} = \frac{\rho_{block}}{\rho_{oil}} = \frac{650}{900} = 0.72$$

$$V_{\text{oil}} = 0.72V_{\text{block}}$$